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Lab1: Implementation of Source code management using git commands
through github

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OBJECTIVE:

Implementation of Source Code Management Using Git Commands Through GitHub.

AIM

To understand and implement Source Code Management (SCM) using Git and GitHub for tracking changes, managing versions, and collaborating on software projects.

THEORY

Source Code Management (SCM) is the process of tracking and controlling changes in software source code. SCM helps developers work collaboratively, maintain version history, and recover previous versions when necessary.

Git is a distributed version control system that enables developers to record changes to files and coordinate work among multiple developers.

GitHub is a cloud-based hosting platform that provides Git repository management along with collaboration tools such as pull requests, issue tracking, and project management.

The combination of Git and GitHub is widely used in Agile Software Development because it facilitates teamwork, continuous integration, version tracking, and code review.

OBJECTIVES

- Learn the basics of Git.
- Create and manage Git repositories.
- Track project changes using commits.
- Connect local repositories with GitHub.

- Pushandpullprojectupdates.
- Understandcollaborativesoftwaredevelopment.

ALGORITHM

1. InstallGitonthe system.
2. CreateaGitHubaccount.
3. CreatearepositoryonGitHub.
4. InitializealocalGit repository.
5. Addproject files.
6. StagefilesusingGit.
7. Commitchanges.
8. LinklocalrepositorytoGitHub.
9. Pushfilesto GitHub.
10. Verifysuccessfulsynchronization.

CODE

- InitializeRepository

```
gitinit
```

- ConfigureUserInformation

```
gitconfig--globaluser.name"yourname"
```

```
gitconfig--globaluser.email"your@email.com"
```

- CheckRepositoryStatus

```
gitstatus
```

- AddFiles

```
gitadd.
```

- CommitChanges

```
gitcommit-m"InitialCommit"
```

- ConnectRepositorytoGitHub

```
gitremoteaddoriginhttps://github.com/username/repository.git
```

- PushtoGitHub

```
gitbranch-M main
```

```
gitpush-uoriginmain
```

- PullLatestChanges

```
gitpulloriginmain
```

OUTPUT

TheGitrepositorywassuccessfullyinitializedandconnectedtoGitHub.

- Repositorycreatedsuccessfully.
- FiletrackedusingGit.
- Changescommittedsuccessfully.
- Remoterepositorylinked.
- ProjectpushedtoGitHub.

RESULT

The implementation of Source Code Management using Git and GitHub was completed successfully. Version control operations such as repository creation, staging, committing, and synchronization with GitHub were performed successfully.

CONCLUSION

This experiment provided practical knowledge of Source Code Management using Git and GitHub. It demonstrated how version control systems help manage software projects efficiently by maintaining code history, enabling collaboration, and supporting Agile Software Development practices.